When the horizontal position of bullet, $x = d$, the time $d = v_{0x}T$. It implies that $T = d / v_{0x}$

At this time T, the vertical distance covered by the bullet is

$$y_b = v_{0y}T - \frac{1}{2}gT^2 = \frac{v_{0y}d}{v_{0x}} - \frac{1}{2}gT^2.$$

$$y_b = \frac{d\,v_0\,sin\theta}{v_0\,cos\theta} - \frac{1}{2}gT^2$$

$$= d\,tan\theta - \frac{1}{2}gT^2 \qquad (3)$$

But from the figure we can write, $tan\theta = \frac{h}{d}$.

$$h = d\,tan\theta\,.$$

By substituting this in the equation (3), we get,

$$y_b = h - \frac{1}{2}gT^2 \qquad (4)$$

At this same time T, the vertical position of the monkey can be calculated from the equation (1)

$$y_m = h - \frac{1}{2}gT^2 \qquad (5)$$

Note that at the time T, the y coordinate of both monkey and bullet is same. It implies that the bullet will hit the monkey.

7. A three storey building of height 100 m is located on Earth and a similar building is also located on Moon. If two persons jump from the top of these buildings on Earth and Moon simultaneously, when they reach the ground, what will be their speed? (g = 10 m s^{-2})

Solution:



For both persons, the Kinematic equations are the same, with u = 0, a_e = g and $a_{moon} = \frac{g}{6}$. Then

$$a_e = g \;and\; a_m = \frac{g}{6}$$

For a person on earth, $V_{earth} = \sqrt{2gh}$

$= \sqrt{2g100} = \sqrt{2\times10\times100}$

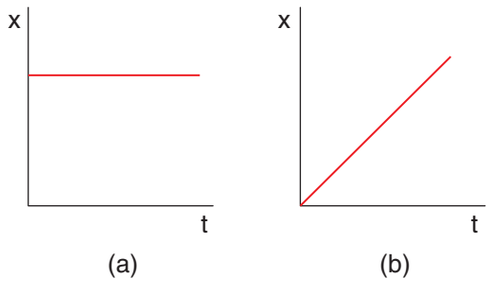
$Hence, V_{earth} = \sqrt{2000}\ m\ s^{-1}$ gives the velocity at the ground, on earth.

Similarly, for a person on the moon,

$$V_{moon} = \sqrt{\frac{2gh}{6}} = \frac{\sqrt{2000}}{\sqrt{6}}\ m\ s^{-1}$$

The person on earth reaches ground with greater velocity than the person on the moon

8. The following graphs represent position – time graphs. Arrange the graphs in ascending order of increasing speed.



(a) (b)

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